

The future-proof nature of Auracast is evident in its compatibility upgrades. High-end smartphones can be easily upgraded to support Auracast, ensuring a seamless transition to this revolutionary technology. Additionally, Auracast bridges facilitate audio sharing with older Bluetooth Classic devices, bridging the gap between technology generations.

Breaking sound barriers in public places

Auracast aims to break down information barriers in public service areas where privacy is not a primary concern. As a result, audio data transmitted via Auracast is not encrypted, allowing receiving devices to access and join the broadcasted content freely. This unencrypted approach ensures that all users in public spaces can easily access audio information, fostering inclusivity.

By extending Bluetooth wireless audio applications beyond the consumer domain to public service areas, Auracast reimagines audio broadcasting. Public spaces now have the potential to provide assistive listening, guided tours, and multilingual support, expanding the traditional boundaries of sound and improving access to audio experiences. Auracast also plays a pivotal role in creating quiet spaces in locations like cinemas, conference centres, and schools for personalised listening.

Recognising the importance of confidentiality in private applications, Auracast offers an optional encryption feature for broadcasted content. Receiving devices in these scenarios can only access the corresponding content by entering a password, mirroring the familiar Wi-Fi user experience. This encryption capability adds an extra layer of security, ensuring that sensitive audio content remains protected and accessible only to authorised users.

Research indicates that approxi-

mately 60 million potential public places worldwide could adopt Auracast technology for service upgrades. This projection anticipates widespread adoption of Auracast-supported receiving devices, including headphones, True Wireless Stereo (TWS) earbuds, and hearing aids in the consumer market. The growing demand for enhanced audio experiences in public spaces, combined with the potential benefits offered by Auracast, creates a conducive environment for market growth and sets the stage for transforming our interaction with the world.

Developing Auracast-enabled devices that resonate with consumers requires careful consideration of the system on chip (SoC). Telink's TLSR9 series audio SoCs stand out as best-in-class chips designed with progress in mind. These chips facilitate seamless integration of existing Bluetooth technologies, ensuring compliance with industry standards. By choosing Telink's SoCs, developers can accelerate development, ensure compatibility, and unlock the full potential of Auracast broadcast audio in their devices, delivering an exceptional audio quality experience to users.

Changing Bluetooth sound with innovative solutions

NXP introduces the NXH3675, a Bluetooth 5.3 LE audio certified chip that not only facilitates Auracast but also supports multi-stream audio with features like broadcast, scan, and delegate for seamless audio sharing. This is designed for efficient functionality with low power consumption, represents a significant leap forward in achieving ultra-low latency applications.

Complementing the NXH3675, NXP's Bluetooth 5.3 LE Audio certified NXH2004 evaluation kit takes Auracast to new heights. With ultra-low power and ultra-low latency capabilities, this evaluation

kit becomes a crucial component in the development process, promising enhanced Bluetooth audio experiences.

Nordic Semiconductor enters the scene with the nRF5340 audio development kit (DK), a versatile platform for Bluetooth LE Audio products. This comprehensive kit includes all necessary components to kickstart development, supporting the complete range of Auracast features. Its configurability allows it to seamlessly operate as a USB dongle for audio data exchange with a PC or function as a business headset or TWS earbud. In various use-cases, the recommendation involves utilising two or more DKs, showcasing its adaptability and versatility in diverse audio applications.

Meanwhile, Cambridge software startup AudioTelligence is making waves in assistive audio technology. Their reference design for the assistive hearing device, Orsana, serves as a foundation for hearing aid and consumer electronics companies to develop their own personalised listening solutions. Designed to seamlessly integrate with Auracast, Orsana facilitates multiple connections to one device, promising a personalised audio stream for each user. This innovation marks a significant step towards transforming the landscape of personalised audio experiences in assistive technologies.

Auracast is emerging like the superhero of Bluetooth audio, swooping in to completely change how we vibe with sound. And the best part? No complicated setups. Just a smooth journey with searching, scanning, and tapping. As we ride this wave of innovation, Auracast is right there, leading the charge into a future where sound connects us all in the most extraordinary ways. **EFY**

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